What is an SLA?

A contract

- Company and external provider, or
- departments within a company

To define

- level/scope of services to be delivered
- metrics by which that service is measured
- remedies or penalties for contract breach

Why are SLAs important?

Clear lines of demarcation

- Expectations of the client and provider are clearly defined

Contract enforcement

- The agreement can be legally enforceable
- If a dispute arises, the parties can refer to the contract to help find a solution

Performance Metrics

- Key performance indicators (KPIs) that can measured and monitored
  - Network Availability
  - Network Performance
  - Support Service

The CIA Triad

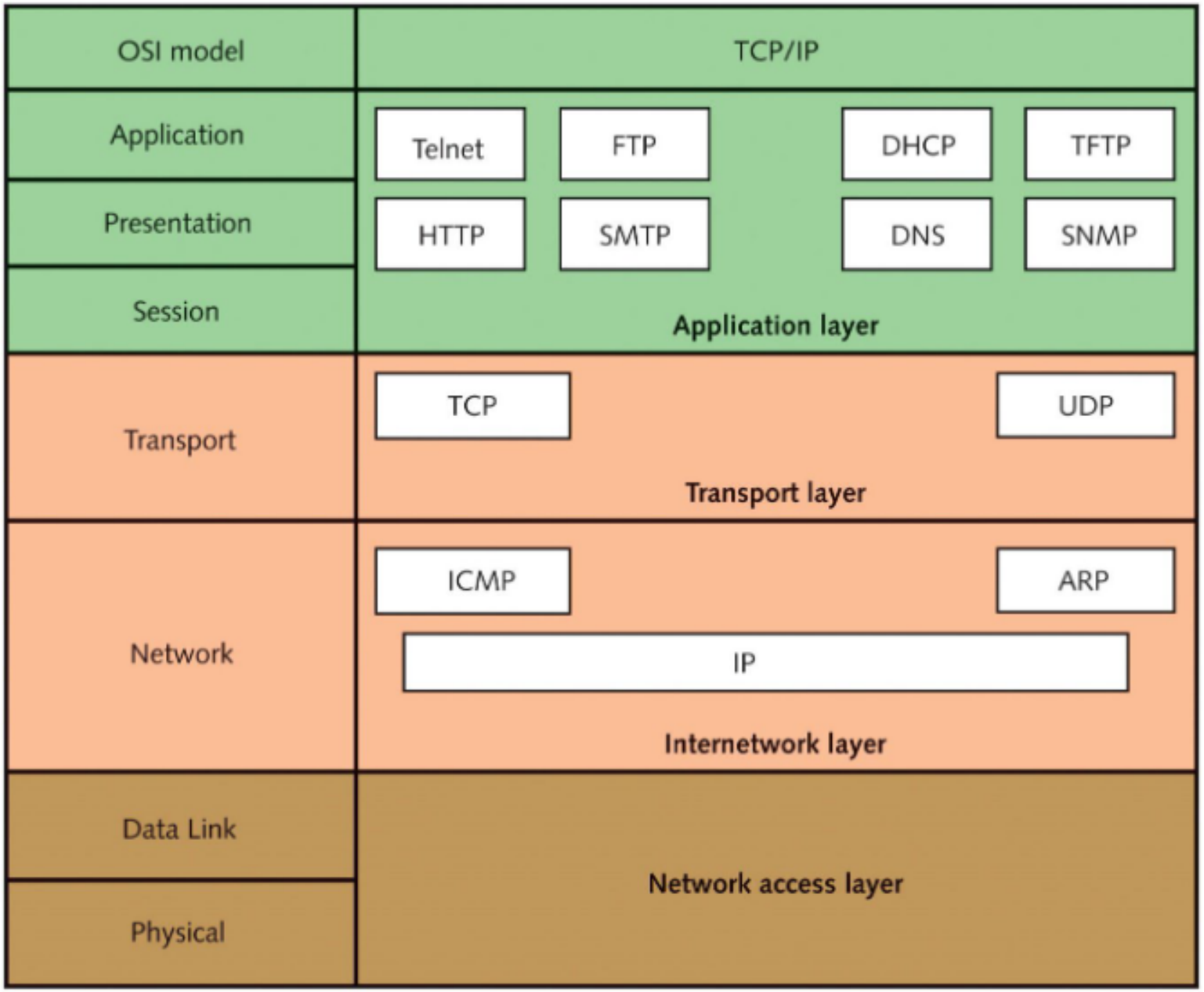
- **Confidentiality**
  - Ensuring that the protection of information assets and networks from unauthorized users
- **Integrity**
  - Ensuring that the modification of information assets is managed in an authorized manner
- **Availability**
  - Ensuring continuous access to information assets and networks by authorized users

Determining Elements of a Network Security Policy (1 of 2)

- Access control policy
  - Specifies how and when users are allowed to access network resources
- Privacy policy
  - Describes what staff, customers, and business partners can expect for monitoring and reporting network use
- Acceptable use policy
  - what purposes network resources can be used
  - what constitutes proper or improper use of network resources

- A network security policy is a document that describes the rules governing access to a company's information resources
- Authentication and authorization enable administrators to control who has access to the network and what users can do on the network
- Securing access to data includes authentication and authorization, encryption, VPNs, security devices like firewall, IDS/IPS
- To protect against threats from external networks with firewalls, IDSs/IPSs implemented on the network perimeter

OSI and TCP/IP models



Protocol Threats

- IP
  - IP Spoofing attack
  - Fragmentation attack
- ARP
  - ARP Cache-poisoning attack
  - MITM attack
- ICMP
  - Redirect attack
  - Ping of death attack
- TCP
  - Predicting TCP Sequences
  - TCP SYN Flood attack
- DHCP
  - DHCP Spoofing
  - Starvation attack
- DNS
  - DNS Cache Poisoning
  - DNS Flood attack

- IoT technologies are often used to underpin developments of smart cities
- IoT Communication Technologies
  - Long range
    - Operating frequency below 1 GHz (sub-GHz) can extend the coverage.
    - With lower-frequency signals, they penetrate buildings and obstacles better than higher frequencies.
  - Short range
    - Most of short-range technologies operate in the 2.4/5 GHz bands.
- 3 models of cloud computing operations:
  - Infrastructure as a Service (IaaS)
  - Platform as a Service (PaaS)
  - Software as a Service (SaaS)